

Magnetism and Matter

Single Correct Type Questions

1. The magnetic moments associated with two closely wound circular coils A and B of radius $r_A = 10$ cm and $r_B = 20$ cm respectively are equal if: (Where N_A , I_A and N_B , I_B are number of turn and current of A and B respectively)

[30 Jan, 2023 (Shift-I)]

- (a) $2N_AI_A = N_BI_B$
 (b) $N_A = 2N_B$
 (c) $N_AI_A = 4N_BI_B$
 (d) $4N_AI_A = N_BI_B$

2. The free space inside a current carrying toroid is filled with a material of relative permeability 1.02. The percentage increase in the value of magnetic field inside the toroid will be

[11 April, 2023 (Shift-I)]

- (a) 2% (b) 0.2%
 (c) 0.1% (d) 1%

3. A rod with circular cross-section area 2 cm^2 and length 40 cm is wound uniformly with 400 turns of an insulated wire. If a current of 0.4 A flows in the wire windings, the total magnetic flux produced inside windings is $4\pi \times 10^{-6}$ Wb. The relative permeability of the rod is

(Given: Permeability of vacuum $\mu_0 = 4\pi \times 10^{-7}\text{NA}^{-2}$)

[31 Jan, 2023 (Shift-I)]

- (a) 12.5 (b) $\frac{32}{5}$
 (c) 125 (d) $\frac{5}{16}$

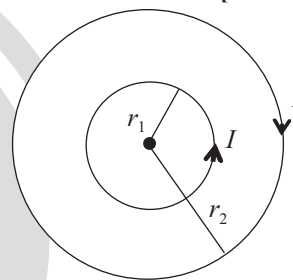
4. A bar magnet having a magnetic moment of $2.0 \times 10^5\text{ JT}^{-1}$, is placed along the direction of uniform magnetic field of magnitude $B = 14 \times 10^{-5}\text{ T}$. The work done in rotating the magnet slowly through 60° from the direction of field is:

[26 June, 2022 (Shift-II)]

- (a) 14 J (b) 8.4 J
 (c) 4 J (d) 1.4 J

5. Two concentric circular loops of radii $r_1 = 30\text{cm}$ and $r_2 = 50\text{cm}$ are placed in $X - Y$ plane as shown in the figure. A current $I = 7\text{A}$ is flowing through them in the direction as shown in figure. The net magnetic momentum of this system of two circular loops is approximately:

[26 July, 2022 (Shift-II)]



- (a) $\frac{7}{2}\hat{k}\text{ Am}^2$
 (b) $-\frac{7}{2}\hat{k}\text{ Am}^2$
 (c) $7\hat{k}\text{ Am}^2$
 (d) $-7\hat{k}\text{ Am}^2$

6. Which of the following statements are correct?

[18 March, 2021 (Shift-II)]

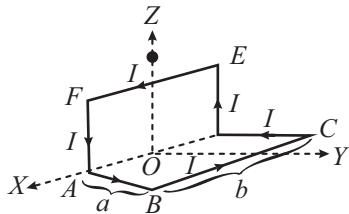
- (A) Electric monopoles do not exist whereas magnetic monopoles exist.
 (B) Magnetic field lines due to a solenoid at its ends and outside cannot be completely straight and confined.
 (C) Magnetic field lines are completely confined within a toroid.
 (D) Magnetic field lines inside a bar magnet are not parallel.

Choose the correct answer from the options given below:

- (a) (A) and (B) only
 (b) (B) and (C) only
 (c) (C) and (E) only
 (d) (B) and (D) only

7. A wire carrying current I is bent in the shape $ABCDEF$ as shown, where rectangle $ABCD$ and $ADEF$ are perpendicular to each other. If the sides of the rectangles are of lengths a and b , then the magnitude and direction of magnetic moment of the loop $ABCDEF$ is

[2 Sep, 2020 (Shift-II)]



- (a) abl , along $\left(\frac{\hat{j}}{\sqrt{5}} + \frac{2\hat{k}}{\sqrt{5}}\right)$
- (b) $\sqrt{2}abl$, along $\left(\frac{\hat{j}}{\sqrt{2}} + \frac{\hat{k}}{\sqrt{2}}\right)$
- (c) abl , along $\left(\frac{\hat{j}}{\sqrt{2}} + \frac{\hat{k}}{\sqrt{2}}\right)$
- (d) $\sqrt{2}abl$, along $\left(\frac{\hat{j}}{\sqrt{5}} + \frac{2\hat{k}}{\sqrt{5}}\right)$
8. A charged particle going around in a circle can be considered to be a current loop. A particle of mass m carrying charge q is moving in a plane with speed v under the influence of magnetic field $\vec{B}$. The magnetic moment of this moving particle is [6 Sep, 2020 (Shift-II)]
- (a) $-\frac{mv^2\vec{B}}{2B^2}$ (b) $\frac{mv^2\vec{B}}{2B^2}$
- (c) $-\frac{mv^2\vec{B}}{2\pi B^2}$ (d) $-\frac{mv^2\vec{B}}{B^2}$
9. Consider a circular coil of wire carrying constant current I , forming a magnetic dipole. [7 Jan, 2020 (Shift-I)]

The magnetic flux through an infinite plane that contains the circular coil and excluding the circular coil area is given by ϕ . The magnetic flux through the area is given by ϕ_0 .

Which of the following is correct?

- (a) $\phi_i = -\phi_0$ (b) $\phi_i > \phi_0$
- (c) $\phi_i = \phi_0$ (d) $\phi_i < \phi_0$
10. A bar magnet having a magnetic moment 5.0 Am^2 is placed in parallel position relative to a magnetic field of 0.4 T . The amount of work done in turning the magnet from parallel to antiparallel position relative to the field direction is [31 Jan, 2023 (Shift-I)]
- (a) 4 J (b) 1 J
- (c) 2 J (d) Zero

11. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

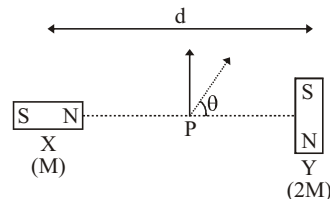
Assertion A: A bar magnet dropped through a metallic cylindrical pipe takes more time to come down compared to a non-magnetic bar with same geometry and mass.

Reason R: For the magnetic bar, Eddy currents are produced in the metallic pipe which oppose the motion of the magnetic bar.

In the light of the above statements, choose the correct answer from the options given below

[11 April, 2023 (Shift-II)]

- (a) Both A and R are true but R is not the correct explanation of A
- (b) A is true but R is false
- (c) Both A and R are true and R is the correct explanation of A
- (d) A is false but R is true
12. A bar magnet is released from rest along the axis of a very long vertical copper tube. After some time the magnet will [10 April, 2023 (Shift-II)]
- (a) Move down with almost constant speed
- (b) Oscillate inside the tube
- (c) Move down with an acceleration greater than g
- (d) Move down with an acceleration equal to g
13. A small bar magnet placed with its axis at 30° with an external field of 0.06 T experiences a torque of 0.018 Nm . The minimum work required to rotate it from its stable to unstable equilibrium position is [4 Sep, 2020 (Shift-I)]
- (a) $9.2 \times 10^{-3} \text{ J}$ (b) $6.4 \times 10^{-2} \text{ J}$
- (c) $7.2 \times 10^{-2} \text{ J}$ (d) $11.7 \times 10^{-3} \text{ J}$
14. Two magnetic dipoles X and Y are placed at a separation d , with their axes perpendicular to each other. The dipole moment of Y is twice that of X. A particle of charge q is passing through their mid-point P, at angle $\theta = 45^\circ$ with the horizontal line as shown in the figure. What would be the magnitude of force on the particle at that instant? (d is much larger than the dimensions of the dipole) [8 April, 2019 (Shift-II)]



- (a) $\sqrt{2} \left(\frac{\mu_0}{4\pi}\right) \frac{M}{(d/2)^3} \times qv$ (b) $\left(\frac{\mu_0}{4\pi}\right) \frac{2M}{(d/2)^3} \times qv$
- (c) $\left(\frac{\mu_0}{4\pi}\right) \frac{M}{(d/2)^3} \times qv$ (d) 0

15. For a moving coil galvanometer, the deflection in the coil is 0.05 rad when a current of 10mA is passed through it. If the torsional constant of the suspension wire is 4.0×10^{-5} , the magnetic field is 0.01 T and the number of turns in the coil is 200, the area of each turn (in cm^2) is:

[25 Jan, 2023 (Shift-II)]

- (a) 2.0 (b) 1.0 (c) 1.5 (d) 0.5

16. The number of turns of the coil of a moving coil galvanometer is increased in order to increase current sensitivity by 50%. The percentage change in voltage sensitivity of the galvanometer will be:

[31 Jan, 2023 (Shift-II)]

- (a) 100% (b) 50%
(c) 75% (d) 0%

17. A hoop and a solid cylinder of same mass and radius are made of a permanent magnetic material with their

magnetic moment parallel to their respective axes. But the magnetic moment of hoop is twice of solid cylinder. They are placed in a uniform magnetic field in such a manner that their magnetic moments make a small angle with the field. If the oscillation periods of hoop and cylinder are T_h and T_c respectively, then: [10 Jan, 2019 (Shift-II)]

- (a) $T_h = T_c$
(b) $T_h = 2T_c$
(c) $T_h = 1.5T_c$
(d) $T_h = 0.5T_c$

Integer Type Questions

18. A wire of length 314 cm carrying current of 14 A is bent to form a circle. The magnetic moment of the coil is _____ Am^2 .
[Given $\pi = 3.14$]

[29 July, 2022 (Shift-I)]



ANSWER KEY

1. (c) 2. (a) 3. (c) 4. (a) 5. (b) 6. (c) 7. (b) 8. (a) 9. (a)
10. (a) 11. (c) 12. (a) 13. (c) 14. (d) 15. (b) 16. (d) 17. (a) 18. [11]

EXPLANATIONS

1. (c) $M = NIA$

$$M_A = M_B$$

$$\therefore N_A I_A A_A = N_B I_B A_B$$

$$\therefore N_A I_A \pi (0.1)^2 = N_B I_B \pi (0.2)^2$$

$$\therefore N_A I_A = 4 N_B I_B$$

2. (a) Given, $\chi_m = 2 \times 10^{-2} = 0.02$

$$B = \mu_r B_0$$

$$B = (1 + \chi_m) B_0 = (1 + 0.02) B_0 = 1.02 B_0$$

Percentage increase in magnetic field

$$= \frac{B - B_0}{B_0} \times 100 = \frac{1.02 B_0 - B_0}{B_0} \times 100 = 2\%$$

3. (c) $\therefore \phi = BA$

$$\therefore \phi = \mu_r \mu_0 \frac{N}{\ell} I \times A$$

$$\text{Thus, } \mu_r = 125$$

4. (a) Work done (W) = change in potential energy (ΔU).

$$W = -MB \cos \theta_2 - (-MB \cos \theta_1)$$

$$= -MB [\cos \theta_2 - \cos \theta_1]$$

$$= -2 \times 10^5 \times 14 \times 10^{-5} (\cos 60^\circ - \cos 0^\circ)$$

$$= -28 \left(\frac{1}{2} - 1 \right) = 14J$$

5. (b) Magnetic momentum, $\vec{M} = IA_1 - IA_2$

$$= I\pi r_1^2 \hat{k} - I\pi r_2^2 \hat{k}$$

$$\Rightarrow \vec{M} = 7 \times \frac{22}{7} \times \left[\left(\frac{30}{100} \right)^2 - \left(\frac{50}{100} \right)^2 \right] \hat{k} \text{ Am}^2$$

$$= 22 \times \left(\frac{-16}{100} \right) \hat{k} \text{ Am}^2$$

$$= -3.52 \hat{k} \text{ Am}^2 \approx -\frac{7}{2} \hat{k} \text{ Am}^2$$

6. (c) (A) Electric monopoles exist while magnetic monopoles do not exist.

(B) Magnetic field lines at the ends and outside of solenoid cannot be confined.

(C) Magnetic field lines are confined within a toroid.

(D) Magnetic field lines inside a bar magnet are parallel.

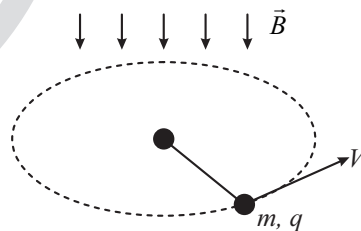
7. (b) $\vec{M} = \vec{M}_1 + \vec{M}_2$

$$\vec{M}_1 = abl \hat{j}$$

$$\vec{M}_2 = abl \hat{k}$$

$$\vec{M} = \sqrt{2} abl \left(\frac{\hat{j}}{\sqrt{2}} + \frac{\hat{k}}{\sqrt{2}} \right)$$

8. (a) $\vec{M} = \frac{1}{2} qvR(-\hat{B})$



$$= \frac{qv}{2} \left(\frac{mv}{qB} \right) \left(\frac{-\vec{B}}{B} \right)$$

$$= -\frac{mv^2}{2B^2} \vec{B}$$

9. (a) As magnetic field lines going inside is equal to the magnetic field lines coming out.

$$\therefore \phi_i = -\phi_o$$

10. (a) $W = -MB (\cos \theta_2 - \cos \theta_1)$

$$W = 2MB = 2 \times 5 \times 0.4 = 4J$$

11. (c) Both A and R are true and R is the correct explanation of A .
12. (a) After some time, the magnet will move with constant speed because of two equal force on magnet.

13. (c) $\tau = MB \sin\theta$

$$0.018 = M \times 0.06 \times 0.5$$

$$\Rightarrow M = 0.6 \text{ Am}^2$$

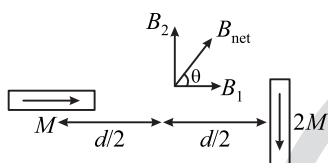
$$W = U_f - U_i$$

$$= MB (\cos\theta_1 - \cos\theta_2)$$

$$= 0.06 \times 0.06 (1 - (-1)) = 7.2 \times 10^{-2} \text{ J}$$

14. (d) For dipole X:

$$B_X = 2 \left(\frac{\mu_0}{4\pi} \right) \frac{M}{(d/2)^3}$$

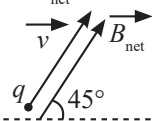


For dipole Y:

$$B_2 = \left(\frac{\mu_0}{4\pi} \right) \frac{2M}{(d/2)^3}$$

$$\therefore B_1 = B_2$$

$\therefore B_{\text{net}}$ is at 45° from the horizontal line.



Velocity of charge q at 45° from the horizontal line.

$$\text{Force on charge, } F = q(\vec{v} \times \vec{B}) = 0$$

15. (b) For the equilibrium of the coil of a moving coil galvanometer, $\tau = K\theta$

where k is torsional constant

$$\Rightarrow NiAB = K\theta$$

$$\Rightarrow A = \frac{K\theta}{NiB} = \frac{4 \times 10^{-5} \times 0.05}{200 \times 10 \times 10^{-3} \times 0.01}$$

$$\Rightarrow A = 1 \text{ cm}^2$$

16. (d) Current sensitivity = voltage sensitivity $\times R$

since current sensitivity is made 1.5 times. Therefore, R also increases 1.5 time.

Hence, voltage sensitivity

$$= \frac{1.5 \text{ current sensitivity}}{1.5 R} = \text{no change}$$

17. (a) $\therefore T = 2\pi \sqrt{\frac{1}{\mu B}}$

$$\frac{T_h}{T_c} = \sqrt{\frac{I_h}{I_c} \times \frac{\mu_c}{\mu_h}} = \sqrt{\frac{2I_c}{I} \times \frac{\mu_c}{2\mu_c}} = 1$$

$$\therefore T_h = T_c$$

18. [11] $\ell = 2\pi r$

$$\Rightarrow 314 \text{ cm} = 2 \times 3.14 r$$

$$\Rightarrow r = 0.5 \text{ m}$$

$$\Rightarrow \mu = iA$$

$$= 14 \times 3.14 \times \frac{1}{4} = 11$$